10.20 域 Field

FCK K/F : K、F是域. F是K的子域

Field Extension ↓ K　F ↘ base field

K is F- linear space [K:F] dimension

Prime Field / Ground Field …最小的域 $\begin{cases} \mathbb{F}_p = \mathbb{Z}/p\mathbb{Z} & char = p \\ \mathbb{Q} & char = 0 \end{cases}$

char：多少个1相加起来为0

Field extension

① $F[x]/(f)$ irreducible poly f　② $Frac(F[x])$　③ K/F and $u \in K$ $F(u)$ K/F(u)/F

$F[x]/(f) \cong F(\alpha)$ if there exists K/F $\alpha \in K$ and $f(\alpha)=0$

Pf. $\varphi: f \longmapsto f(x)$

Ker φ $\varphi: F[x] \to F(\alpha)$

$Ker\varphi = (g)$

$f \in Ker f$ $g \mid f$

any homomorphism $\phi: F \to F'$ Fields　ϕ is injective or ϕ is trivial 所有映射到0

Ker为理想（域上理想是0/整）

K/F $\alpha \in K$

1) α is algebraic $\exists f \neq 0 \in F[x]$ $f(x)=0$ 有限维的域

2) α is transcendental $1, \alpha, \alpha^2, \alpha^3 \dots$ are F- linear indep

$Frac(F(x)) \cong F(\alpha)$ $\frac{f}{g} \mapsto \frac{f(\alpha)}{g(\alpha)}$

尺规



作不出$(\sqrt[3]{2}, 0)$

可做 $Q \subset Q(\sqrt{2}) \subset Q(\sqrt[4]{2})$

$[F:Q] = 2^d$ 假如可以，则 $F/Q[\sqrt[3]{2}]/Q$

$[F:Q] = [F \cdot Q[\sqrt[3]{2}]] \cdot [Q[\sqrt[3]{2}]:Q] = 3$

证 H/K/F

$[H:F] = [K:F][H:K]$

Splitting Field 分裂域

Splic 分裂

f split in K means $f(x)=\prod_{i=1}^{\deg f}(x+u_i)$

⇑

$F[x]$ K/F $u_1,\dots u_n,\dots\in K$

Spliting Field of f = $F(u_1,u_2,\dots)$

1) Spliting Field Exists $[K:F]\le(\deg f)!$

$f\in F[x]$ irreducible g s.t. $g\mid f$ and $\deg g>1$

$p_1 p_2\cdots p_t$

F 扩张到 $F[y]/(g(y))$ 扩域中 y 本身就是 g(y) 根

$f\in(F(x)/(g(y))][x]$ f 次数有限，扩域中不可约的依链使重复，至多有限次扩张就完全分裂

x^3-2 = $\mathbb{Q}[x]$ $\mathbb{Q}[y]/(y^3-2)\cong\mathbb{Q}[\sqrt[3]{2}]$

$x^3-2=(x-\sqrt[3]{2})(x^2+\sqrt[3]{2}x+\sqrt[3]{4})$ — $\mathbb{Q}[\sqrt{2},\sqrt[3]{2}(-1)^{\frac{2}{3}}]$ 经6次扩张

2) Spliting Field Unique (在同构意义下)

K/F is a splitting field of f

pick irreducible $g\mid f$ and $\deg g>1$

pick irreducible $g_1\mid f$ (in $F[x]$) and $g_1\mid f$

$F_2=F_1[y]/(g_1)\cong F(\alpha,\alpha_1)$ (K)

$F_1=F[y]/(g)\cong F(\alpha)$

F

F is finite field $|F|=p^d$

$x^{p^d}-x\in F_p[x]$ Splitting Field $K/\mathbb{F}_p$ of $x^{p^d}-x$

$x^{p^d}-x=\pi(x-u_i)$ $u_i\in K$ 因为导数和原式互素，则 u_i 两两不同.

$\{u\in K\mid u^{p^d}-u=0\}$

a, b $(a+b)^{p^d}-(a+b)=a^{p^d}+b^{p^d}-a-b=0$ a+b，ab都在集合当中，大小为 p 的 d 次方

Let $f\in\mathbb{F}_p[x]$ is a deg deg-d irreducible poly in $\mathbb{F}_p[x]$

$K=\mathbb{F}_p[x]/(f)$ $\exists\alpha\in K$, $f(\alpha)=0$ $x-\alpha\mid f$ $x-\alpha\mid x^{p^d}-x$

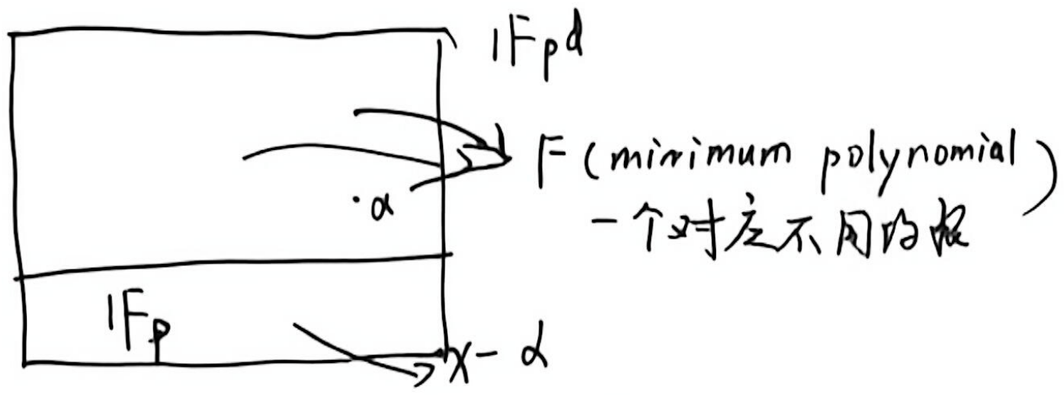
$x-\alpha\mid\gcd(f,x^{p^d}-x)$ 作 gcd 时仍在 $\mathbb{F}_p[x]$ 里 $\gcd(f,x^{p^d}-x)\in F_p[x]$ (不为1)

$f=\gcd(f,x^{p^d}-x)$ $f\mid x^{p^d}-x$

Enumerate $\alpha\in\mathbb{F}_{p^d}$ "minimal polynomial of α" $f(\alpha)=0$ $f\neq0$ f has minimal degree 极小多项式

$\Rightarrow$ f is irreducible

$\mathbb{F}_p[x]/(f)\cong\mathbb{F}_p(\alpha)$ $\mathbb{F}_{p^d}/\mathbb{F}_p(\alpha)/\mathbb{F}_p$ $\deg f=[\mathbb{F}_p(\alpha):\mathbb{F}_p]\mid d$



$x^{p^d}-x$

π ∥ f

$f \in \mathbb{F}_p[x]$

f ineducible. and deg f | d

$f \mid x^p - x$

f monic：最高次项为1

$\varphi_d = \prod_{f \in \mathbb{F}_p[x]} f$ ineducible & monic, $\deg f = d$

$\deg \varphi_1 = p$

$\deg \varphi_2 = p^2 - p$

$\deg \varphi_3 = p^3 - p$

$\deg \varphi_6 = p^6 - p^3 - p^2 + p$

$x^{p^6} - x = \varphi_6 + \varphi_3 + \varphi_2 + \varphi_1$

$\deg \varphi_d \geq p^d - \sum_{\substack{d' \mid d \\ d' \neq d}} p^{d'}$

$x^p - x = \varphi_1$

$x^{p^2} - x = \varphi_1 \varphi_2$

$x^{p^d} - x = \prod_{d' \mid d} \varphi_{d'}$

$p^d = \sum_{d'} \deg(\varphi_p)$

$= \geq$

$\mathbb{F}_q[x]$ $q = p^d$ Input $f \in \mathbb{F}_q[x]$ output Factorization

① square free if $g^2 \mid f$ $g \mid \gcd(f, f')$

$x^{2p} - x^p + 3 = (x^2 + x + 3)^p$

② distinct-degree $f = r_1 r_2 r_3 \cdots$ $\deg r_1 \neq \deg r_2$

$\deg r_1 < \deg r_2$ $r_1 \mid x^{q^d} - x$ $r_2 \nmid x^{q^d} - x$ $\gcd(x^{q^d} - x, f)$

gcd 是于为一个非平凡的分解)

③ some degree $f = r_1 r_2 r_3$ $\mathbb{F}[x]/(f)$

r_1, r_2, r_3 degree = d

$\mathbb{F}_q[x]/(f) \cong \mathbb{F}_{q^d}[x]/(r_1) \times \mathbb{F}_q[x]/(r_2) \times \mathbb{F}_q[x]/(r_3)$

$= \mathbb{F}_{q^d}$

暂且只考虑 odd q

Pick random $s \in \mathbb{F}_q[x]/(f)$

$\gcd(s^{\frac{q^d-1}{2}} - 1, f)$

得于的非平凡的因式分解

(s_1, s_2, s_3)

y^{q^d} ∥ y

any $y \neq 0$

$y^{\frac{q^d-1}{2}} = \pm 1$ $y^{q^d-1} = 1$